

Smart Gardens Engineering Design Brief

Need Statement:

In recent years, there has been an increase in the number of rural and urban regions that are considered to be “food deserts,” low-income neighborhoods with limited access to healthy and affordable foods that typically affect minority residents disproportionately¹. In response, many community members have looked to develop their own small-scale gardens – in their backyards or greenhouses, on balconies, apartment rooftops, or small community garden plots. However, with rain patterns becoming more unpredictable throughout the year and longer stretches of drought in summer months, residents have struggled to maintain these gardens effectively.

You work for an environmental engineering consulting firm that is trying to get hired to develop an automated irrigation system to more effectively manage water use for these kinds of local gardens. One particular client is Tesla STEM High School, whose Environmental Club is looking to develop a school garden to help beautify the outdoor space and highlight plant species that are native to the region. Before implementing the final system in the field, your team must build a proof-of-concept model to demonstrate to the client.

Requirements:

Based on the client's requests, your team must be able to meet the following key deliverables:

1. Research **one or more native plants** to use for model garden testing, including target values for key environmental metrics (light, air temperature, humidity, soil temperature, soil moisture)
2. Develop an **Arduino-based system** that can:
 - a. Collect environmental sensor data for surrounding light, air temperature, humidity, soil temperature, and soil moisture;
 - b. Use sensor data to control a small water pump to provide optimal water flow to plants and soil;
 - c. Display live sensor data, labeled and easy to read/interpret while system is running;
 - d. Store all sensor data locally (through Excel or an embedded memory card) or remotely (through a wireless network card)
3. Analyze stored sensor data and create **data visualizations** to assess plant/soil health and provide recommended environmental metrics
4. Design (via **3D models**) and construct **physical structures** that can:
 - a. Contain soil, plant(s) and water reservoir(s), allowing for easy portability;
 - b. Store all electronics (Arduino/sensors/pump) while protecting them from external damage and allowing access for repairs/troubleshooting;
 - c. Manage irrigation of water from reservoir(s) to plants/soil, including drainage collection as needed
5. Assess the **overall sustainability** of the system, including:
 - a. Progress and opportunities toward net-zero water, energy, and greenhouse gas emissions
 - b. Potential benefits and applications relevant to environmental justice-focused organizations and initiatives

¹ https://www.ers.usda.gov/webdocs/publications/45014/30940_err140.pdf

Resources, Roles, and Responsibilities:

Your firm's manager has allocated 4.0 – 5.0 FTE (full-time employees), 3.5 weeks of work time, and a small supplemental budget to meet all the client's deliverables.

Team members' general roles and responsibilities are summarized below (full list of responsibilities and deliverables are given at the end of this briefing). If your team has five members, each member will take on a 1.0 FTE role; if your team has four members, one member will take on a 0.5 FTE role for both Lead Researcher and Lead Data Analyst.

- Project Manager (1.0 FTE)
- Lead Programmer (1.0 FTE)
- Lead Structural Designer (1.0 FTE)
- Lead Researcher (0.5 – 1.0 FTE)
- Lead Data Analyst (0.5 – 1.0 FTE)

Your team may be able to request some additional materials beyond those initially provided. Any material requests will need to be submitted and approved by your manager, with accurate links, descriptions, quantities, and pricing information included. You should expect a lead time of up to two (2) business days for all email requests, and your team is responsible for understanding lead times for materials to arrive.

Timeline, Deliverables, and Documentation:

At the end of 3.5 weeks, your manager is expecting a **fully functional prototype** to share with the client and a **detailed presentation** of the work done by all team members to justify the budget used on this project.

Your manager is expecting your team to **thoroughly document all work** done on this project, including meeting and research notes, preliminary and final versions of all code and structural designs, photos of key progress points in the project, and a detailed work schedule and budget for your team. All team files should be stored together and organized in an easily accessible way throughout the project.

Due dates for your team's overall deliverables are given below. **Your team is responsible for setting and managing intermediate timelines and checkpoints to ensure all key performance indicators (KPIs) listed below are met for each role.** Each team member should be prepared to share documentation and periodic updates of their progress as requested by your manager.

Functional Model Due Date: Tu 6/9

Presentation Due Date: M 6/15

Role	Primary Responsibilities	Deliverables / Key Performance Indicators (KPIs)
Project Manager (1.0 FTE)	• Collects regular status reports from all team leads and communicates overall progress to management (Mr. Bonomo). • Ensures project timeline and deliverables are met for each subsystem and helps troubleshoot problems / allocate resources as needed. • Uses project management tools (RACI chart, Gantt chart, Budget Sheet, and others) to communicate and measure progress of key deliverables. • Delegates documentation, reporting, and other tasks to team leads as needed. • Creates and/or delegates work to create final presentation (PowerPoint) and recorded demo video of final functioning system. • Coordinates live demo of final system.	• Completed Weekly Scorecards for each week of project • Updated RACI Chart • Updated Gantt Chart • Final Project Budget • Dated notes to track meetings / overall team progress throughout the project • Final PowerPoint presentation with clear and concise summaries of all team work • Recorded video demonstrating full functionalities of final system • Single folder of all team documents, organized by subsystem
Lead Programmer (1.0 FTE)	• Researches and understands all hardware (Arduino, sensors, pump) and software (Arduino IDE, Excel, etc.) required for system operation. • Writes/edits code to operate input and output hardware. Works closely with Lead Researcher to incorporate correct environmental metrics related to your plant. • Works closely with Lead Data Analyst to ensure all required data is collected, displayed, and stored accurately. • Reports regular schedule/status updates to Project Manager as needed. • <i>[OPTIONAL: Writes/edits code to incorporate local data storage using microSD adapter and/or remote data storage using wireless adapter.]</i>	• Preliminary annotated code files for each individual sensor (light, air temp/humidity, soil temp, soil moisture, relay, push button) • Final annotated code file for combined system • Detailed photos of hardware setup (Arduino, sensors, pump, etc.) • Dated notes to track meetings/subsystem progress throughout the project
Lead Structural Designer (1.0 FTE)	• Researches and understands a variety of options for garden structure, irrigation methods, and electronics management and is able to communicate benefits and trade-offs of each. • Creates concept sketches for at least three (3) design options for each key structural component (plant/soil/water storage, irrigation method, and electronics management). • Documents and provides justification for final design choices, working closely with Lead Researcher and Lead Programmer to ensure all plant and sensor needs are able to be met. • Creates 3D CAD design of final structural system. • Constructs/acquires materials for final structural components and assembles them into final system. • Reports regular schedule/budget/status updates to Project Manager as needed.	• Concept Sketches (3+ for each of plant/soil/water storage, irrigation, electronics storage) • Documentation of final design choices • 3D CAD files for final system design • Detailed photos of key structural components (garden storage, irrigation, electronics storage) and final assembled system • Dated notes to track meetings/subsystem progress throughout the project

Lead Researcher (0.5 – 1.0 FTE)	• Researches similar systems online and provides examples/links to other subsystem leads as needed. • Researches and understands at least three (3) plant options for garden model, including key needs / environmental targets for light, air temp, humidity, soil temp, soil moisture, etc. • Documents final plant selection and responsible for obtaining plants/soil used in garden model. • Works closely with Lead Programmer to incorporate correct env. metrics into final code. • Works closely with Lead Structural Designer to ensure structure meets all plant needs. • Researches opportunities to achieve net-zero water, energy, and greenhouse gas emissions documents progress for overall system. • Researches and understands relevant issues of environmental justice, food justice, and food security in rural and urban contexts and identifies related organizations and initiatives locally. • Reports regular schedule/status updates to Project Manager as needed. ----- Include below items for 1.0 FTE ----- • Researches and understands B-Corp Certification and provides recommendations for how this project can help the client to obtain it. • Assesses the overall sustainability of each subsystem and provides recommendations / opportunities for improvements in the future.	• Research notes, diagrams, and comparison table of key health metrics for 3+ plant options • Documentation and detailed photos of final plant selection • Research Briefings (a.k.a. summary documents) for system progress and opportunities around: ○ Net-zero water, energy, and GHG emissions for overall system ○ Issues, organizations, and initiatives around environmental justice, food justice, and food security ----Include below for 1.0 FTE---- ○ B-Corp Certification for client ○ Sustainability audit of each subsystem with improvement opportunities
Lead Data Analyst (0.5 – 1.0 FTE)	• Researches and understands a variety of data visualization techniques relevant to system data and justifies decisions for chosen methods. • Works closely with Lead Programmer to integrate hardware/software for data collection, ensuring all required data is collected, displayed, and stored accurately. • Manages Excel spreadsheets and/or other files to read and store all sensor data. • Creates compelling and easy-to-read data visualizations to analyze stored sensor data. • Uses data to assess plant/soil health and provide recommended environmental metrics for your team's plants/soil. • Reports regular schedule/status updates to Project Manager as needed. ----- Include below items for 1.0 FTE ----- • Researches possible artificial intelligence models, such as lobe.ai for analyzing plant health through captured images. • Creates a sample model of an artificial intelligence platform such as lobe.ai to demonstrate AI's role in assessing plant health through captured images.	• Research notes, links, and example data visualizations relevant to system data with explanation for final choices • Raw data files and graphs (Excel or similar) for 5 key sensor metrics (light, air temp, humidity, soil temp, soil moisture) • Data visualizations (different from raw data graphs) for analyzed/compiled data for 5 key metrics • Data Summary Report (Word doc or similar) that includes final system data, formatted graphs/visuals, recommended plant metrics, and written summaries of all info provided ----Include below for 1.0 FTE---- • Research notes, folder of source images, and any other files needed to demonstrate a sample model of an AI platform to assess plant health

Team Roster:

Period: _____ Team: _____ Firm Name: _____

Enter first and last names of all team members in their chosen roles below. Each team member must choose one Team Lead role and one Primary Support role.

Role	Team Lead	Primary Support
Project Manager		
Lead Programmer		
Lead Structural Designer		
Lead Researcher		
Lead Data Analyst		